

[SIN-WI-3516_A] Work Instruction Re-flashing Ray 1 Boards

Author(s)	Title
Rollin Luo	Senior Operations Associate
Reviewer	Title
Cerys Cook	Chief People and Operations Officer

Revision History

Revision	Changes	Author (Title)	Approved by (Title)	Date	Effective Date
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1.0 Introduction

The Swift Ray 1 device flashing process is used to apply the applicable device firmware onto the Swift Ray 1 driver board. Swift Ray 1 devices that were previously flashed may need to be re-flashed for various reasons, which may include firmware version updates or device troubleshooting. This work instruction will describe the process for re-flashing assembled Swift Ray 1 devices.

1.1 Scope

This work instruction applies to Swift Ray 1 devices that have previously been flashed and assembled fully but now require re-flashing.

2.0 Applicable and Reference Documents

2.1 Applicable Standards

Document Title	Document #
N/A	N/A

2.2 Reference Documents

Document Title	Document #
Work Instruction: Board-Flashing Procedure for Ray 1	[SIN-WI-2213]
Work Instruction: Ray 1 Rework Process	[SIN-WI-2880]
Prod Ops - Swift Ray 1 Inventory Assembled Devices (Device History Record)	[SIN-DHR-2152]
Supplies and Tool Log and Inventory	[QMS-REC-2239]
Equipment Log	[QMS-REC-0106]
Rework Log	[QMS-REC-2990]
Device Integrated Test Procedure	[SIN-WI-2151]
Quarantine Product Log	[QMS-REC-2166]
Work Instruction: ESD Wrist Strap Verification	[QMS-WI-2600]
SOP: Control of Non-Conforming Product	[QMS-SOP-0033_F]

3.0 Training Required

To perform this procedure, operators should be trained on this document, as well as all work instructions listed in the above table.

4.0 Terms and Definitions

Term	Meaning
Flashing/Re-flashing	Uploading the firmware onto the Ray 1 board

4.1 Applicable Acronyms

Acronym	Definition
DHR	Device History Record

ESD	Electro-Static Discharge
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5.0 Process Roles

Role	Responsibility
Operations Associate/Operator	Performs procedure in Work Instruction, ensuring all aspects are documented as described by the procedure.
Operations Manager/Regulatory	Reviews and approves completed documentation and procedures.

6.0 Supplies/Equipment Required

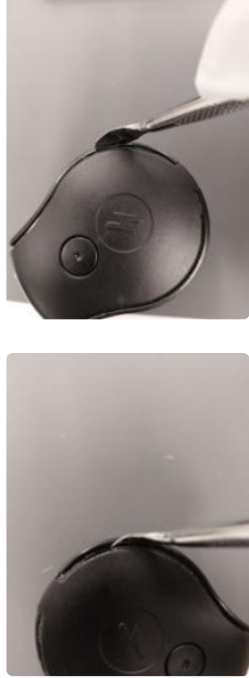
The following supplies/equipment should be prepared prior to beginning this procedure.

Supply/Equipment Number	Supply/Equipment Name
SUP-C016	Spudger/Pry Tool
SUP-C021	Devboard with connector cable (5-colour)
EQUIP-017A	Production Laptop
EQUIP-027 A/B/C/D	ESD Wrist Strap

7.0 Work Instructions for Re-flashing Devices

Re-flashing assembled Swift Ray 1 devices is considered a REWORK process. Prior to beginning the process of re-flashing any assembled devices, operators shall review [\[SIN-WI-2880\]](#) Work Instruction: Ray 1 Rework Process, linked in Section 3.0 of this document, in conjunction with the instructions provided in this Work Instruction. The Ray 1 Rework Process Work Instruction defines the required rework activities and shall be followed to ensure that all applicable actions are completed.

Additionally, many of the required steps for re-flashing are consistent with those described in [\[SIN-WI-2213\]](#) Work Instruction: Board-Flashing Procedure for Ray 1. Where this Work Instruction references SIN-WI-2213, operators shall follow the applicable instructions provided in that Work Instruction. Operators should review SIN-WI-2213 before beginning the re-flashing process to familiarize themselves with the applicable board-flashing procedure.

Step	Task	Notes
1.	Gather the devices that will be re-flashed. As you gather each device for re-flashing, update the status of the device in the Device History Record (DHR) [SIN-DHR-2152] to "Rework" to indicate that a rework process is currently being performed on the device.	Ensure that the assigned serial number of each device is noted and associated properly with the board number in the DHR, as after the re-flashing process is complete the serial number will need to be re-provisioned. The provisioning process occurs during the retesting of the device, which is described in Section 6.0 of this document. Each device should keep the same serial number that was assigned previously when it is re-provisioned.
2.	Follow Steps 3 to 12 outlined in [SIN-WI-2213] to set up the appropriate files on the production laptop for the re-flashing of the devices.	Setting up the programs and files on the production laptop for re-flashing processes is the same as if you were flashing the Swift Ray 1 board for the first time. If the re-flashing is being performed due to a firmware update, ensure that the version number on the file name is the correct version.
3.	Prior to proceeding with the rest of this procedure, the operator should ensure that their ESD wrist strap is worn and verified as per [QMS-WI-2600] . The ESD wrist strap should be worn whenever the operator may come into direct contact with the internal components of the device. Take a device being re-flashed and open the back cover using a spudger. The spudger should be inserted between the connection of the front cover and main body, near one of the protruding wedges as shown in the images. There are four wedges located on this side of the device. Take care not to use too much force when opening the cover as	

Step	Task	Notes
	excessive force may result in chipping or scratches. Only the driver board needs to be directly connected for the flashing process, so you do not need to open the front cover of the device.	NOTE: The device may be either turned on or off during the beginning of this process; the initial status does not make a difference as the device will automatically shut off once re-flashing begins.
4.	Follow Steps 13 to 17 in [SIN-WI-2213] to perform the re-flashing of the Swift Ray 1. Upon successful flashing, the blue LED light should begin blinking every three seconds, as described in Step 17. If this does not occur, an error may have disrupted the process, such as: 1. Software reports an error (error messages appear in red text) 2. The connection between the production laptop and the board was disrupted 3. The previous firmware on the board was not erased correctly. 4. There is a defect with one of the components on the board that prevents it from operating. Regardless of the type of error that may have occurred, the operator should carefully re-start the process from beginning to end (Step 13 to 17), ensuring that every step is repeated in order. Do not start the process partway through, as this may result in further errors occurring.	NOTE ON REPEATING THE RE-FLASHING PROCESS: If Step 17 does not occur as expected, the operator should make three further attempts to perform Steps 13 to 17 in [SIN-WI-2213] as described in this procedure. Should the repeated attempts to perform these steps continue to not result in a successful flashing process, the operator should label the device as "Re-flashing failed" and set it aside in a separated section before proceeding. The operator should refer to the guidance in Step 8 of this procedure regarding unsuccessful re-flashing once they have completed their work.

Step	Task	Notes
5.	Once the re-flashing is complete, replace the back cover of the device. The back cover should slide down onto the rails at the spine of the housing. Press the snaps into place on either side and work around to the other side near the charge port. Inspect the exterior of the device once the cover is replaced, ensuring all snaps are seated, no gaps are noticed, no external damage such as scratches or cracks are observed, and the charging port is aligned correctly.	NOTE: If the exterior inspection shows any issues as described, the device should be set aside and labelled with the type of damage that has occurred. The operator should refer to the guidance in Step 8 of this procedure regarding device damage.
6.	If the firmware version of the device was updated due to re-flashing, find which row the re-flashed device is listed in [SIN-DHR-2152] and update Column K to reflect the firmware version that is now on the device. Log the re-flashed device in the Rework Log [QMS-REC-2990] , referencing this work instruction for the procedure. Set the re-flashed device aside in a separate area temporarily, ensuring that it does not get mixed up with other devices in the inventory with different firmware versions. Proper storage of devices with updated firmware will be described in Section 6.0.	
7.	Repeat Steps 3-6 for each device that needs to be re-flashed.	
8.	Once all the devices have been re-flashed, update the status (Column D) of all successfully	Once a decision has been made on any devices that were not successfully re-flashed, operators

Step	Task	Notes
	re-flashed devices in [SIN-DHR-2152] to “Needs QC” as they will need to be re-tested. If any devices were not successfully re-flashed (per the procedure outlined in Step 4) or device damage is noticed (per Step 5), they should remain in a “Rework” status in [SIN-DHR-2152] temporarily. The Operations Manager and Regulatory team will be informed of any unsuccessful re-flashings, and the team will collectively reach a decision on how the device should be handled (ex. Begin another rework process, move to Quarantine, etc.)	must ensure to update: [SIN-DHR-2152] DHR – ensure that the status is appropriately updated. [QMS-REC-2166] Quarantine Product Log – if the device is moved to Quarantine, enter it into the Quarantine Product Log as a new line item [QMS-REC-2990] Rework Log – enter the device as a new line item into the Rework Log, indicating clearly in Column J “Comment” if the device was moved into Quarantine, needs to be reworked again, or any other different action occurred. If another rework action needs to be performed for any device, it should be entered again as its own line item in the Rework Log, with a clear indication of the details of rework.

8.0 Storage and Re-testing of Re-flashed Devices

Any devices that have been re-flashed shall undergo the QC process again to verify that the devices continue to function as intended before they can be re-approved for market release. Once the operator has completed the re-flashing steps outlined in Section 5.0, the devices shall be stored and re-tested in accordance to the instructions below. This process is intended to prevent storage mix-ups and ensure that all required re-testing and rework activities are appropriately documented.

Step	Task	Notes
1.	Place all re-flashed devices in the Pre-QC Inventory section, marked with a yellow sticker. The devices should be grouped based on their: 1. Firmware version number 2. Date of re-flashing All devices that are kept together in the same group should be identical for both of the above characteristics.	

Step	Task	Notes
	Both the firmware version number and date of re-flashing should be clearly written on a label (either on the Pre-QC yellow sticker if there is room, or on a separate label stored with the group of devices).	
2.	In the Device Integrated Test Procedure section of the DMS [SIN-WI-2151] create a separate folder for re-testing the re-flashed devices. The folder's name should clearly indicate: 1. The date in a clear format (YYYY-MM-DD) 2. Firmware version number currently on the devices 3. Brief description of the reason for re-flashing the devices (e.g. "Firmware Update" or "Device Error") Follow the Device Integrated Test Procedure to re-QC all the devices, storing the test records in the folder created above. During the provisioning step of the re-QC, ensure that the SAME serial number that was previously used for the device is provisioned again.	If the device does not pass the test procedure due to an issue, or for any reason does not need to be or cannot be re-tested by the Operations team, enter the device into the Rework Log [QMS-REC-2990] and write a note indicating the reason the device is not being re-tested, alongside the rest of the required information in each column. The entry will be reviewed and signed off/approved by the Operations Manager or Regulatory team with the date of approval. Update the device status in the DHR [SIN-DHR-2152] appropriately based on what action will be taken (e.g. move to Quarantine, place back in Rework, etc.) Any device placed into Quarantine should be entered into the Quarantine Product Log [QMS-REC-2166] as a new line item.
3.	Once a device has passed QC again, it can be placed in the Post-QC Inventory section and marked with a green sticker. Similarly to the Pre-QC Inventory guidance in step 1, any devices with an updated firmware version should be kept separately from devices that are still on a previous firmware version and labelled clearly to indicate which firmware version number has been flashed onto the device.	

Step	Task	Notes
4.	Enter all devices into the Rework Log [QMS-REC-2990] , filling in the information in each column. It is mandatory to provide the information for each column, except for Column J "Comment", which may be labelled "N/A" if no relevant comment is required. The information provided in the entry should provide a clear explanation of the reason why the device is re-flashed. Once the reworks have been approved, update the status of all Passed devices in the DHR [SIN-DHR-2152] to "Available".	The entries will be reviewed and signed off/approved by the Operations Manager or Regulatory team with the date of approval. Devices are authorized to be restored to the Available inventory once their entry in the Rework Log is approved.

Page Attachments